

M.S. Student, Biomedical Engineering
Brain-Computer Interface Lab (Prof. Sung-Phil Kim)
Ulsan National Institute of Science and Technology (UNIST)

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RESEARCH INTERESTS

Brain-Computer / Brain-Machine Interfaces · Neural Encoding & Decoding of Vision and Motor Control · Cognitive Neuroscience of Sensory Perception · Brain Imaging

SKILLS

TOOLS MATLAB · Python · EEGLAB · SPM · FSL · MySQL

METHODS EEG · fMRI · Eye-tracking

EDUCATION

2025 - Current	M.S. in Biomedical Engineering Brain-Computer Interface Lab (Advisor: Prof. Sung-Phil Kim) · UNIST, Ulsan, South Korea
2019 - 2025	B.S. in Biomedical Engineering Minor in Electrical Engineering · UNIST, Ulsan, South Korea
2023 - 2024	Republic of Korea Air Force – Tactical Air Control Party Air Support Operations Center · Mandatory Military Service
2016 - 2018	Hansung Science High School Seoul, South Korea

RESEARCH EXPERIENCE

2020 - Current	Brain-Computer Interface Lab, UNIST Advisor: Prof. Sung-Phil Kim ■ Led human neurophysiology studies using EEG (and EEG + eye-tracking) with end-to-end responsibility from experimental design to analysis (MATLAB / EEGLAB; Python as needed). ■ Key contributions across projects: paradigm development and implementation, participant recruitment and session running, preprocessing / feature extraction, and interpretation / reporting of results.
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SELECTED PROJECTS

- **EEG responses of emotion across OTT subscription fees** (*joint project with Korea University*) — Designed and implemented the experimental paradigm; conducted EEG experiments with 10+ participants; analyzed EEG data to characterize emotional differences across cost conditions.
- **Neural basis of the choice-overload effect** — Developed and executed a complex EEG + eye-tracking paradigm; conducted experiments with 50+ participants; processed and analyzed large-scale datasets to probe neural processes under different preference conditions.
- **fMRI study on tactile illusion** — Conducted 3T fMRI experiments with 30+ participants; analyzed BOLD signals using MATLAB / SPM / FSL; compared EEG vs. fMRI methodologies for sensory-mechanism studies.
- **EEG signal analysis** — Signal processing and analysis using EEGLAB and MATLAB.

Spring 2022

Decision Neuroscience & Cognitive Engineering Lab, UNIST

Advisor: Prof. Dongil Chung

- **Resting-state EEG analysis of drug-induced parkinsonism and Parkinson's disease** (*joint project with Dongtan Sacred Heart Hospital*) — Analyzed resting-state EEG from PD and DIP patients; performed spectral analyses and EEG-clinical correlation analyses to identify differentiating neural signatures and support diagnostic insight.

Summer 2020

Bio and Neural Interfaces Lab, KAIST

Advisor: Prof. Seongjun Park

- **Mechanical interaction of thermally drawn fiber implants with brain tissue** — Investigated tissue-device interactions for fiber-based neural implants using in vivo surgery, histology, and computational modeling. Performed mouse-brain implant surgeries with fiber-based neural probes; verified immune responses via immunohistochemistry; conducted finite-element analysis to model probe-tissue mechanics.

PUBLICATIONS

- 2026 **S. Seo**, S. Lee, N. Lee & S.-P. Kim. *Reducing cognitive demand fails to mitigate choice overload: multimodal neural and gaze evidence for effort conservation*. **iScience** (preprint: [bioRxiv](#), 2026-02). **ACCEPTED IN PRINCIPLE**
- 2024 **S. Seo**, S. Kim, S.-P. Kim, J.-H. Kim, S. Y. Kang & D. Chung. *Low-frequency EEG power and coherence differ between drug-induced parkinsonism and Parkinson's disease*. **Clinical Neurophysiology**, 168, 131–138.
- 2023 J. Park, S. Wang, S. Lee, **S. Seo**, N. Lee & S. Kim. *Viewer emotional response to webtoon-based drama: an EEG analysis*. **International Journal of Human-Computer Interaction**, 1–15.
- 2021 K. Kim, C. Sung, J. Lee, J. Won, W. Jeon, **S. Seo**, K. Yoon & S. Park. *Computational and histological analyses for investigating mechanical interaction of thermally drawn fiber implants with brain tissue*. **Micromachines**, 12(4), 394.
- 2021 S. Lee, **S. Seo**, N. Lee & S.-P. Kim. *The effect of OTT subscription fee on content engagement*. **The Journal of Korean Institute of Communications and Information Sciences**, 46(3), 572–582.

CONFERENCE PRESENTATIONS

- Nov 2026 **S. Seo**, O.-S. Kwon & S.-P. Kim. *Closed-loop motor-imagery brain-computer interface training reshapes visual-motion perception*. **Society for Neuroscience (SfN) 2026**, Washington, DC. **POSTER**
- Jul 2026 **S. Seo**, O.-S. Kwon & S.-P. Kim. *EEG neurometric-psychometric dissociation in visual motion perception*. **Brain Engineering Society of Korea 2026**, Jeju, South Korea. **TALK**
- May 2026 **S. Seo**, O.-S. Kwon & S.-P. Kim. *EEG neurometric function mirrors psychometric performance in visual motion perception*. **Vision Sciences Society (VSS) 2026**, St. Pete Beach, FL. **POSTER**
- Feb 2026 J. Kim, C.-E. Yoon, **S. Seo** & S.-P. Kim. *Real-time single-trial decoding in a calibration-free P300-based BCI using a compact eight-channel EEG configuration*. **14th International Conference on Brain-Computer Interface (BCI) 2026**, Jeongseon, South Korea. **POSTER**
- Sep 2024 **S. Seo**, S. Kim, S.-P. Kim, J. Kim, D. Chung & S. Y. Kang. *Quantitative EEG analysis for comparing drug-induced parkinsonism and Parkinson's disease*. **International Congress of Parkinson's Disease and Movement Disorders 2024**, Philadelphia, PA. **POSTER**

- Nov 2022 J.-H. Kim, J. Kim, **S. Seo** & S.-P. Kim. *Somatosensory cortical representations of assimilation effects by vibrotactile stimulation: an fMRI study*. **Society for Neuroscience (SfN) 2022**, San Diego, CA. [POSTER](#)
- Sep 2022 **S. Seo**, S. Lee, N. Lee & S.-P. Kim. *Preference modulates the effect of choice overload with less involvement of working memory*. **International Congress of Physiological Anthropology 2022**, Eugene, OR. [POSTER](#)
- May 2022 S. Lee*, **S. Seo***, N. Lee & S.-P. Kim. *Attenuated working memory accounts for choice overload in highly preferred alternatives*. **25th Annual Meeting of the Korean Society for Brain and Neural Sciences**, Incheon, South Korea. [POSTER](#)

* Equal contribution.

RESEARCH GRANTS

- 2026.03 – Current **Augmentation Technology for Overcoming the Limits and Disorders of Human Cognition and Intelligence**
Graduate Researcher
Information Technology and Telecommunications Planning & Evaluation Institute (IITP) · 2024.07 – 2031.12
- 2025.09 – Current **Music BrainCoder: A Non-Invasive BCI for Reproducing Imagined Music and Enhancing Musical Performance**
Graduate Researcher
National Research Foundation of Korea (NRF) · 2025.03 – 2030.02
- 2022.08 – 2023.02 **Future Brain Convergence Technology Development Project**
Undergraduate Researcher
"Development of brain-inspired computational models of active and adaptive intelligence for AI applications."
Ministry of Science and ICT (MSIT) · 2019.06 – 2023.12
- 2022.04 – 2023.02 **Brain Science Proprietary Technology Development Project**
Undergraduate Researcher
"Development of analysis methods for ultra-high-resolution neuroimaging signals and information-processing models of neural networks." Ministry of Science and ICT (MSIT) · 2019.06 – 2023.12
- 2022.03 – 2022.07 **Smart Healthcare Research Center**
Undergraduate Researcher
- 2020.07 – 2022.02 **Smart Media Services Innovation Convergence Project**
Undergraduate Researcher
"Development of neuro-artificial-intelligence convergence technology for next-generation OTT service."
Ministry of Science and ICT (MSIT) · 2020.07 – 2027.12

HONORS & AWARDS

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| 2022 | Excellent Research Award, 2nd Place
Department of Biomedical Engineering Faculty, UNIST, South Korea |
| 2021 | Student Creative Research Project, 1st Place
Smart Media Service Research Center, Korea University, South Korea |
| 2020 | Outstanding Grades for the Semester
Article 62 of the School Regulations · UNIST, South Korea |
| 2020 | Student Creative Research Project, 2nd Place
Smart Media Service Research Center, Korea University, South Korea |
| 2019 | Outstanding Grades for the Semester
Article 62 of the School Regulations · UNIST, South Korea |
| 2019 – Current | National Science & Engineering Undergraduate Scholarship
Korea Student Aid Foundation (Korean Government) · Full scholarship |

TEACHING EXPERIENCE

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| Spring 2026 | Machine Learning Methods for Neuroengineering
Teaching Assistant · UNIST |
| Fall 2025 | Brain & Cognitive Engineering
Teaching Assistant · UNIST |

SERVICES & OTHER PROFESSIONAL ACTIVITIES

PEER REVIEW

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| 2026 | Journal of Neural Engineering
Co-reviewer (with Prof. Sung-Phil Kim) · IOP Publishing |
| 2026 | Frontiers in Network Physiology
Reviewer (ad hoc) |

UNIVERSITY & DEPARTMENTAL SERVICE

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| 2022 | Undergraduate Student Council, UNIST
Election Campaign · Emergency Response Committee |
| 2021 | College of Information & Biotechnology Student Council, UNIST
Division of Academic Studies, School of Electrical Engineering |
| 2020 | School of Biomedical Engineering Student Council, UNIST
Preparation Task Force |
| 2020 | School of Electrical & Computer Science Engineering Student Council, UNIST
Emergency Response Committee · Planning Department |
| 2020 | Campus Assistant, Freshman Support Center, UNIST
Mentor for 18 first-year students (1 year) |
| 2020 | School of Human Factors Engineering Student Council, UNIST
Planning Department |